

Docker - Communication with containers

```
● ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/docker/
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image pull elasticsearch:7.2.0
7.2.0: Pulling from library/elasticsearch
8ba884070f61: Pull complete
2211b14f8b24: Pull complete
617ccdb47f3d: Pull complete
915ee6b2c338: Pull complete
b414b7f29a7d: Pull complete
547bfdd35d62: Pull complete
8353a2ed248c: Pull complete
Digest: sha256:84b5bc2fd15b0f1f5bf78c8c6ee34b6ae5a46ab81be1c2cfa678eea0c6457a46
Status: Downloaded newer image for elasticsearch:7.2.0
docker.io/library/elasticsearch:7.2.0
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --rm -e "discovery.type=single-node" -p 9200:9200 --name my_es_container elasticsearch:7.2.0
2eeae981d0b87036a37fb3bb2384629968cc7ddd1554406744d821ab5f9133c2
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ curl -X GET -i http://localhost:9200
HTTP/1.1 200 OK
content-type: application/json; charset=UTF-8
content-length: 508

{
  "name" : "2eeae981d0b8",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "Y5KWx22gS76thQRn_sBBUg",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
```

Open URL <http://localhost:9200/>

```
{
  "name": "2eeae981d0b8",
  "cluster_name": "docker-cluster",
  "cluster_uuid": "Y5KWx22gS76thQRn_sBBUg",
  "version": {
    "number": "7.2.0",
    "build_flavor": "default",
    "build_type": "docker",
    "build_hash": "508c38a",
    "build_date": "2019-06-20T15:54:18.811730Z",
    "build_snapshot": false,
    "lucene_version": "8.0.0",
    "minimum_wire_compatibility_version": "6.8.0",
    "minimum_index_compatibility_version": "6.0.0-beta1"
  },
  "tagline": "You Know, for Search"
}
```

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container inspect my_es_container | grep IPAddress
      "SecondaryIPAddresses": null,
      "IPAddress": "172.17.0.2",
      "IPAddress": "172.17.0.2",
```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ curl -X GET -i http://172.17.0.2:9200
HTTP/1.1 200 OK
content-type: application/json; charset=UTF-8
content-length: 508

{
  "name" : "2eeae981d0b8",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "Y5KWx22gS76thQRn_sBBUg",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop my_es_container
my_es_container
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --rm\
> -e "discovery.type=single-node"\
> -p 9201:9200 -p 9301:9300\
> --name my_es_container elasticsearch:7.2.0
d63860bdbc82d4e024e5978616e78f91ae89b093eb5c83145750cd21c5b3d3c4
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ curl -X GET -i http://127.0.0.1:9201
HTTP/1.1 200 OK
content-type: application/json; charset=UTF-8
content-length: 508

{
  "name" : "d63860bdbc82",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "0-KgKQ0JSEGytCSHtpGH9A",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container ls

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
		NAMES			
d63860bdbc82	elasticsearch:7.2.0	"/usr/local/bin/dock_..."	About a minute ago	Up About a minute	0.0.0.0:9201->9200/tcp, :::9201->9200/tcp,
0.0.0.0:9301->9300/tcp, :::9301->9300/tcp		my_es_container			
d6cc0b428d28	postgres:16-alpine	"docker-entrypoint.s..."	6 weeks ago	Up 34 minutes	0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
		pg_container			
475951b12750	dpape/pgadmin4	"/entrypoint.sh"	6 weeks ago	Up 34 minutes	443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80
/tcp		pgadmin4_container			

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --rm\
> -e "discovery.type=single-node"\
> -p 9201:9200 -p 9301:9300\
> --name my_es_container2 elasticsearch:7.2.0
402778bb54a62854c1da160fc24c4922fdded97540ecaa18c88b42655928fc92
docker: Error response from daemon: driver failed programming external connectivity on endpoint my_es_container2 (d11e6d55cf647348e6398ffdd097039
edc0b340bc62b987f175fa2b9cdad3066): Bind for 0.0.0.0:9301 failed: port is already allocated.
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image pull datascientest/curl:latest
latest: Pulling from datascientest/curl
21c83c524219: Pull complete
54949da4944b: Pull complete
Digest: sha256:65c0eab9e3e924a1d13e3bc29a2b47957f6dade2d3110edae83f72ecf9dccc0
Status: Downloaded newer image for datascientest/curl:latest
docker.io/datascientest/curl:latest
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm datascientest/curl:latest http://example.org

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed

<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style type="text/css">
    body {
      background-color: #f0f0f2;
      margin: 0;
      padding: 0;
      font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    div {
      width: 600px;
      margin: 5em auto;
      padding: 2em;
      background-color: #fdfdff;
      border-radius: 0.5em;
      box-shadow: 2px 3px 7px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
      color: #38488f;
      text-decoration: none;
    }
    @media (max-width: 700px) {
      div {
        margin: 0 auto;
        width: auto;
      }
    }
  </style>
</head>

<body>
<div>
  <h1>Example Domain</h1>
  <p>This domain is for use in illustrative examples in documents. You may use this
  domain in literature without prior coordination or asking for permission.</p>
  <p><a href="https://www.iana.org/domains/example">More information...</a></p>
</div>
</body>
</html>
100 1256 100 1256 0 0 4534 0 --:--:-- --:--:-- --:--:-- 4534
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm datascientest/curl:latest http://127.0.0.1:9200

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed

curl: (7) Failed to connect to 127.0.0.1 port 9200: Connection refused
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm datascientest/curl:latest http://127.0.0.1:9201

% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed

curl: (7) Failed to connect to 127.0.0.1 port 9201: Connection refused
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ # if the container address is not the same on your machine, change it
● scientest/curl:latest http://172.17.0.2:9200ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --rm datascient
est/curl:latest http://172.17.0.2:9200
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed

0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
{
  "name": "d63860bdbc82",
  "cluster_name": "docker-cluster",
  "cluster_uuid": "0-KgKQJSEgytCSHtpGH9A",
  "version": {
    "number": "7.2.0",
    "build_flavor": "default",
    "build_type": "docker",
    "build_hash": "508c38a",
    "build_date": "2019-06-20T15:54:18.811730Z",
    "build_snapshot": false,
    "lucene_version": "8.0.0",
    "minimum_wire_compatibility_version": "6.8.0",
    "minimum_index_compatibility_version": "6.0.0-beta1"
  },
  "tagline": "You Know, for Search"
}
100 508 100 508 0 0 33866 0 --:--:-- --:--:-- --:--:-- 33866
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
cea2c69b3bc9        bridge              bridge              local
2ef515e83bb9        host                host                local
7181dfed741e        none                null                local
3421b4023b34        sql-relational-db-implementation_default bridge              local
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "cea2c69b3bc92513b0513da5ddd5cacfc1985437a88dbd88098b943858d1ec97",
    "Created": "2025-06-14T19:33:52.677895437Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "d63860bdbbc82d4e024e5978616e78f91ae89b093eb5c83145750cd21c5b3d3c4": {
        "Name": "my_es_container",
        "EndpointID": "9e46406d1784dc5b9fc44b2ce2a9f314b1e969959ec7354128c6f4f461ecc607",
        "MacAddress": "02:42:ac:11:00:02",
        "IPv4Address": "172.17.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {}
  }
]
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network create --subnet 172.50.0.0/16 --gateway 172.50.0.1 my_network
59058e2f80e3122bf02fa50dc831d890e84720786f5b3daa935ffa3171b4ce59
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
cea2c69b3bc9        bridge              bridge              local
2ef515e83bb9        host                host                local
59058e2f80e3        my_network          bridge              local
7181dfed741e        none                null                local
3421b4023b34        sql-relational-db-implementation_default bridge              local
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network inspect my_network
[
  {
    "Name": "my_network",
    "Id": "59058e2f80e3122bf02fa50dc831d890e84720786f5b3daa935ffa3171b4ce59",
    "Created": "2025-06-14T20:24:07.449148037Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.50.0.0/16",
          "Gateway": "172.50.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --rm\
> -e "discovery.type=single-node"\
> --network my_network\
> --name my_es_container3 elasticsearch:7.2.0
9e7a354701801b9b75c05015daab98339a3216d68b2bc5af9d1713975f43df79
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker network inspect my_network
[
  {
    "Name": "my_network",
    "Id": "59058e2f80e3122bf02fa50dc831d890e84720786f5b3daa935ffa3171b4ce59",
    "Created": "2025-06-14T20:24:07.449148037Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.50.0.0/16",
          "Gateway": "172.50.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "9e7a354701801b9b75c05015daab98339a3216d68b2bc5af9d1713975f43df79": {
        "Name": "my_es_container3",
        "EndpointID": "24dd6e37b23e06bfb57591748ebaae7f767c32e16f5bf55a387eb367e93e5a25",
        "MacAddress": "02:42:ac:32:00:02",
        "IPv4Address": "172.50.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run datascientest/curl:latest http://172.50.0.2:9200
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left     Speed
0         0    0     0    0     0      0  0 --:--:-- 0:01:13 --:--:-- 0

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --network my_network datascientest/curl:latest http://172.5
0.0.2:9200
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    508    100    508     0     0    949     0 --:--:-- --:--:-- --:--:--    949
{
  "name" : "9e7a35470180",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "gD2HerVKT4uC_pKjCvTNA",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run --network my_network datascientest/curl:latest http://my_es
_container3:9200
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    508    100    508     0     0   2170     0 --:--:-- --:--:-- --:--:--   2170
{
  "name" : "9e7a35470180",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "gD2HerVKT4uC_pKjCvTNA",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container stop my_es_container
my_es_container
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker container run -d --rm \
> -e "discovery.type=single-node" \
> --network host \
> --name my_es_host_container elasticsearch:7.2.0
12e268e20fdf405836d79acf44c5eb9405b4ef3d57e531f906e2ac2c651a3055
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

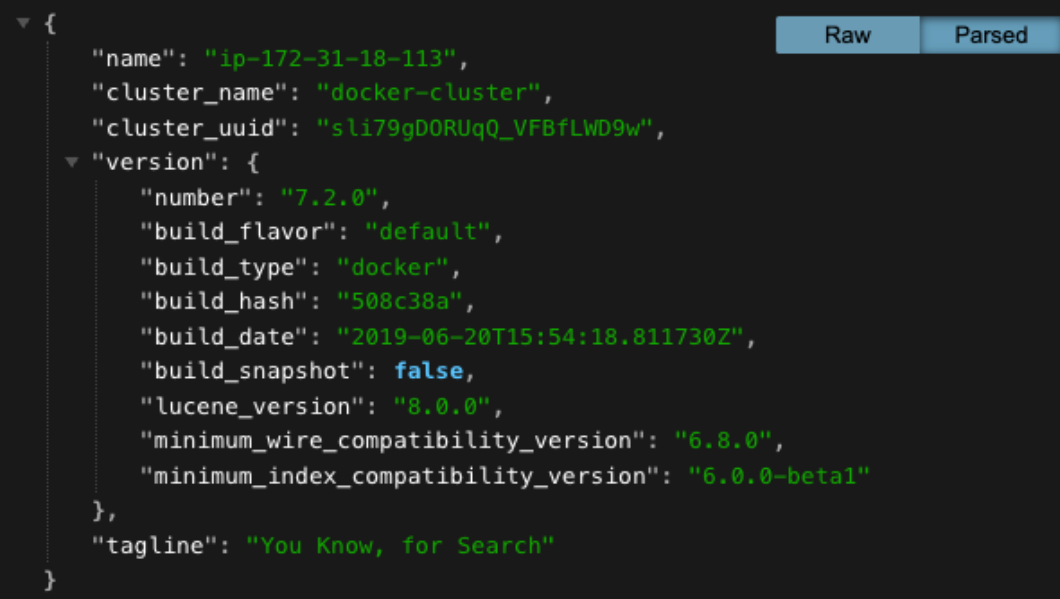
```

● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ curl -X GET -i http://127.0.0.1:9200
HTTP/1.1 200 OK
content-type: application/json; charset=UTF-8
content-length: 512

{
  "name" : "ip-172-31-18-113",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "sli79gDORUqQ_VFBfLWD9w",
  "version" : {
    "number" : "7.2.0",
    "build_flavor" : "default",
    "build_type" : "docker",
    "build_hash" : "508c38a",
    "build_date" : "2019-06-20T15:54:18.811730Z",
    "build_snapshot" : false,
    "lucene_version" : "8.0.0",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$

```

Open URL <http://127.0.0.1:9200/>



The screenshot shows a web browser window with a dark background. At the top right, there are two tabs: "Raw" (selected) and "Parsed". The main content area displays a JSON object representing the Elasticsearch cluster health. The JSON is formatted with syntax highlighting: strings are in green, booleans in blue, and numbers in green. The structure is as follows:

```
{
  "name": "ip-172-31-18-113",
  "cluster_name": "docker-cluster",
  "cluster_uuid": "sli79gD0RUqQ_VFBfLWD9w",
  "version": {
    "number": "7.2.0",
    "build_flavor": "default",
    "build_type": "docker",
    "build_hash": "508c38a",
    "build_date": "2019-06-20T15:54:18.811730Z",
    "build_snapshot": false,
    "lucene_version": "8.0.0",
    "minimum_wire_compatibility_version": "6.8.0",
    "minimum_index_compatibility_version": "6.0.0-beta1"
  },
  "tagline": "You Know, for Search"
}
```